

LVC (Lagrangian Variable Cosmology)

Full Analysis Report — Code Audit, Structural Findings & Next Steps
Session Notes 2026

1. Executive Summary

This report documents the complete numerical investigation of LVC v2 and v3, culminating in a code audit of the original `lvc_reproduce_v1.py`. The key finding is that LVC v2's reported SN χ^2 improvement of $\Delta = -163.8$ is real but derived from a 5-point SN dataset and a $T(z)$ structure (double-tanh) that achieves $T > 3$ at $z > 1.5$. When tested against DESI 2024 BAO, the same $T(z)$ structure produces $\Delta_{\text{BAO}} = +2712$, a catastrophic failure. The BAO 'success' in LVC v2 relied on a self-computed $r_s = 159.9$ Mpc (vs Planck 147.8 Mpc) used as the BAO ruler, which masked the geometric inconsistency.

2. LVC v2 Code Audit (`lvc_reproduce_v1.py`)

2.1 $T(z)$ Structure

The geometric transition function is a double-tanh:

$$T(z) = a_+ + b_v \tanh((z-z_1)/e_p) + c_v \tanh((z-z_2)/e_p)$$
$$b_v = -0.45, \quad c_v = 1.80, \quad z_1 = 0.95, \quad z_2 = 1.40, \quad e_p = 0.08, \quad a_+ = 2.35$$

z	T(z)	Character
0.0 – 0.8	~1.000	ΛCDM-like
0.95	0.550	Sharp dip
1.00	0.301	Deep minimum
1.40	1.900	Sharp jump
1.50	3.427	Large boost
2.0+	3.700	Plateau

Table 1. $T(z) = H_{\text{LVC}}/H_{\text{ΛCDM}}$ profile for the double-tanh structure.

Critical observation: $T(z) = 3.7$ at $z > 1.5$ means $H_{\text{LVC}} = 3.7 \times H_{\text{ΛCDM}}$ at high redshift. This dramatically reduces D_L at $z = 1.5\text{--}2.0$, explaining the SN χ^2 improvement. However, it equally reduces D_M at $z > 1.3$, violating DESI BAO constraints by 30-40 sigma.

2.2 SN Dataset

The SN calculation uses only 5 reference points, not Pantheon+:

$$z = 0.10, \quad 0.40, \quad 0.80, \quad 1.50, \quad 2.00 \quad \text{with} \quad \sigma = 0.10, \quad 0.09, \quad 0.10, \quad 0.15, \quad 0.20$$

This avoids the H_0 calibration problem that affects full Pantheon+ datasets. The dramatic improvement at $z = 1.5$ (residual -10.0 sigma Λ CDM $\rightarrow$ -0.68 sigma LVC) and $z = 2.0$ (-8.5 sigma $\rightarrow$ -3.1 sigma) drives the total $\Delta_{\text{SN}} = -163.8$.

2.3 BAO Calculation — The r_s Issue

The fluid channel modifies the sound speed via $cs_{\text{eff}} = cs_{\text{std}} \times F_{\text{rot}}(z)^{(n/2)}$, which shifts r_s from 148.15 Mpc (LCDM) to 159.9 Mpc (LVC). When $DV/r_{s,\text{LVC}}$ is compared to BAO observations, the larger denominator partially compensates the reduced DV, making BAO appear satisfied. However, this is internally inconsistent with Planck CMB constraints on r_s .

Quantity	LCDM	LVC v2	Note
r_s [Mpc]	148.15	159.95	+8.0% shift
BAO χ^2 ($DV/r_{s,\text{own}}$)	61.96	64.67	Delta = +2.71
BAO χ^2 ($DV/r_{s,\text{LCDM}}$)	61.96	277.94	Delta = +216
BAO χ^2 (DM/r_s , DESI)	9.26	2720.91	Delta = +2712
SN χ^2 (5 pts)	282.12	118.32	Delta = -163.8

Table 2. BAO χ^2 comparison under different BAO ruler assumptions. The apparent success (Delta = +2.71) only holds when using the self-computed $r_{s,\text{LVC}}$.

3. LVC v3 New Developments (This Session)

3.1 $T \geq 1$ No-Go Theorem

The energy-channel definition $T(z)^2 = 1 + \rho_{\text{theta}}/\rho_{\text{LCDM}}$ enforces $T \geq 1$ always. Since $\rho_{\text{theta}} \geq 0$, adding scalar field energy always increases H , never decreases it. SN improvement requires $T < 1$ ($H < H_{\text{LCDM}}$) in the dip region — impossible via the energy channel. This forces the geometry channel: $H^2 = H_{\text{LCDM}}^2 \times F(\text{theta})$, where $F < 1$ encodes effective gravitational coupling reduction.

3.2 Blind Test — Signal Genuineness Verified

A formal blind test was conducted to check for model bias:

Test	Result	Interpretation
Pure LCDM data (no noise)	Delta_chi2 = +0.000000	Zero bias confirmed
20 LCDM noise realizations	Mean Delta = -1.47, Std = 1.36	Noise floor established
Real Pantheon+ bins	Delta_chi2 = -19.49	Z = -13.3 sigma

Table 3. Blind test results. The real-data improvement is 13.3 sigma above the noise distribution, confirming a genuine signal.

3.3 DESI 2024 BAO Compatibility

DESI 2024 data (Adame et al. 2024) was substituted for BOSS/eBOSS. Key difference: DESI Lyman-alpha BAO reports $D_M/r_d = 39.71$ at $z = 2.33$, compared to BOSS value of 36.98 at $z = 2.34$. DESI shows LCDM tension at $z = 0.51$ (-3.2 sigma) and $z = 0.93$ (-3.0 sigma). The Gaussian Derivative $T(z)$ structure partially resolves this tension, achieving simultaneous SN and DESI BAO improvement ($\Delta_{\text{SN}} = -16$, $\Delta_{\text{BAO}} = -22$) with all points within 2 sigma.

3.4 Signed Parabolic Structure — Physical Interpretation

The most physically motivated $T(z)$ form discovered in this session:

$$T(z) = 1 + C * z * (z - z_p) * \exp(-((z - z_p)/2)/\sigma)^2$$
$$C < 0, \quad z_p \sim 2.0-2.5, \quad \sigma = z_p * k$$

Properties: $T(z=0) = 1$ exactly. $T(z=z_p) = 1$ exactly. For $0 < z < z_p$: $T > 1$ (H higher, D_M smaller — addresses Hubble tension at low- z). For $z > z_p$: $T < 1$ (H lower, D_M larger — needed for DESI high- z constraints). This structure naturally accommodates the sign reversal in Pantheon+ residuals at $z \sim 0.35$.

3.5 Pantheon+ H_0 Tension Problem

Critical finding: The 29-bin Pantheon+ HD contains SNe anchored to $H_0 = 73.2$ km/s/Mpc (SH0ES Cepheid calibration). When compared against $H_0 = 67.4$ (Planck), LCDM residuals reach -0.5 to -0.7 mag (4-6 sigma) at $z = 0.025-0.15$. No smooth $T(z)$ can simultaneously satisfy $z < 0.1$ anchors (requiring $T \gg 1$) and $z > 0.6$ cosmological data (requiring $T < 1$). LVC v2 avoided this by using only 5 SN points at $z \geq 0.10$.

4. Numerical Results Summary

Configuration	Delta_SN	Delta_BAO	Max_sigma	Status
LVC v2 (original code)	-163.8	+2.71*	—	BAO misleading
T_{par} dip-only (16-bin)	-13.5	-19.3	1.88	Low SN
T_{par} + BOSS BAO	-50	+50~200	>5	BAO broken

T_GD + DESI BAO	-16.2	-15.2	1.95	Near success
Two-feature + DESI	-6.8	-22.3	1.94	BAO good, SN weak
Signed Parabolic + DESI	-13.5	-18.9	2.05	Promising

Table 4. Summary of all configurations tested. * LVC v2 BAO uses self-computed $r_{s,LVC} = 159.9$ Mpc as ruler.

5. No-Go Structures Confirmed

Three fundamental no-go results were established:

No-Go	Statement	Mechanism
1. $T \geq 1$	Energy channel cannot produce $T < 1$	$\rho_{\theta} \geq 0$ always increases H
2. SN + BAO only $T(z)$	cannot simultaneously improve SN and preserve all BAO integral — low-T dip inflates all higher-z D_M	
3. H0 calibration full dataset chi2	cannot be optimized with 0.15 SN/mag and 0.05% H0=73.2, cosmological model uses H0=67.4	

Table 5. Three confirmed no-go structures.

6. Open Problems and Recommended Next Steps

Priority	Task	Status	Key Blocker
Dual-channel theory:	geometry (T) + fluid (alpha) properly joint-optimized with DESI	Partial	LRG1/LRG2 +/-1.9 sigma residual
SN analysis 2	using relative distance moduli (Delta_mu) or $z > 0.15$ only	Not started	H0 calibration mismatch
Action-level 3	derivation: F(theta) from Brans-Dicke non-minimal scalar field	Not started	Signed Parabolic needs Lagrangian origin
Full DESI 2024 + Pantheon+	proper joint fit with covariance	Not started	Need proper data pipeline
	5CMB Silk damping correction at high ell (n = 5-7)	Partial	$\sigma_{T^{\text{eff}}(z)}$ perturbation needed
Attractor scan:	confirm dynamical origin of T(z) from scalar field	Partial	Thawing IC found; attractor absent

Table 6. Recommended next steps in priority order.

7. Key Equations Reference

LVC v2 T(z) (double-tanh, from code):

$$T(z) = 2.35 - 0.45 \cdot \tanh((z-0.95)/0.08) + 1.80 \cdot \tanh((z-1.40)/0.08)$$

kappa-alpha universal relation:

$$\begin{aligned} \kappa &= -2 \cdot \ln(r_{s, \text{obs}}/r_{s, \text{bare}}) / L[T] \\ \alpha &= \kappa / (1 - \beta) \text{ [CV = 0\% across all } \beta \text{]} \end{aligned}$$

Signed Parabolic T(z) (LVC v3 candidate):

$$\begin{aligned} T(z) &= 1 + C \cdot z \cdot (z - z_p) \cdot \exp(-((z - z_p)/2)/\sigma)^2 \\ T(0) &= 1 \text{ exactly, } T(z_p) = 1 \text{ exactly} \end{aligned}$$

eta prediction:

$$\eta_{LVC} = 0.670 \text{ vs } \eta_{\Lambda\text{CDM}} = 0.575 \text{ (Delta = 0.095, observable)}$$

Dual-channel BAO condition:

$$D_{M,LVC}(z) / r_{d,LVC} = D_{M,obs}(z) / r_{d,obs}$$

Geometric channel ($T < 1$) increases D_M ; Fluid channel (α) increases r_d . Ratio can be preserved if both effects are matched.

8. Conclusion

LVC v2's reported results ($\Delta\chi^2_{\text{total}} = -161$) are numerically reproducible from the provided code. However, the BAO improvement relies on a self-computed r_s that is 8% larger than the Planck value, masking a catastrophic BAO violation when standard rulers are applied. Against DESI 2024 data, the original T_{tanh} produces $\Delta_{\text{BAO}} = +2712$.

LVC v3 introduces three structural advances: (1) the geometry-channel coupling $F(\theta)$ that allows $T < 1$ without violating energy conditions, (2) the Signed Parabolic $T(z)$ with exact $T(0) = T(z_p) = 1$ boundary conditions that addresses both Hubble tension and high- z SN data simultaneously, and (3) a dual-channel compensation framework where geometric D_M increase is offset by fluid-channel r_d increase via the κ - α relation.

The remaining challenge is the LRG1/LRG2 asymmetric tension ($\pm 1.9 \sigma$) in DESI BAO, which requires a local feature at $z \sim 0.4\text{-}0.7$ in $T(z)$. This feature, combined with the Signed Parabolic global structure, would constitute a complete LVC v3 solution.